

J-Lens in Ouro: A Confirmed Late-Pass Band and a Target-Dependent Early-Pass Deficit

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Abstract

Does a Jacobian-based readout reveal intermediate content that a model’s own output head misses during recurrent computation? I compare J-Lens with the raw logit lens in frozen Ouro-2.6B, which applies 48 shared blocks over four recurrent passes. Layerwise discovery analysis found a positive J-Lens region in pass 4 whose location and size five independent calibration fits reproduced. I then froze one estimator (fit01) and that band, physical layers 26–37 of pass 4, before evaluating 160 new two-hop questions with 80 new intermediate concepts. In the band, J-Lens improves excess hit@10 by **23.19 percentage points** (95% dependency-group bootstrap interval **16.29–32.90**); intended-concept recovery is 37.92% against 14.43% for the raw lens. Under the same final-target estimator, the same physical band shows the opposite sign in passes 1–3 (−6.5, −8.6 and −11.1 points; post-confirmation, simultaneous intervals excluding zero), so the pattern is pass-dependent at fixed depth. The late advantage remains positive with within-domain controls (20.8 points, 9.1–32.5) and under an exhaustive tokenized-input audit, FP64 reference scoring and a cross-hardware state replay (changes of at most 0.42 points), but it depends strongly on the estimator: changing the derivative target or the position aggregation substantially reduces it. In a post-confirmation analysis on the same states, initial-study banks targeting each pass’s own exit reverse the deficit: in the band they exceed the final-target lens of their family by 22.6, 26.5 and 34.2 points and the raw lens by 16.4, 18.3 and 23.2 points (passes 1–3; simultaneous intervals exclude zero), so the early deficit belongs to the estimator’s target, not to the passes. The result supports a localized, estimator- and target-specific improvement in concept recovery, not general J-Lens superiority, improved answer accuracy, or causal use of the recovered content.

1 The question

A readout can miss useful content even when that content is present in a model’s hidden state. The raw logit lens applies the model’s output head directly to an intermediate activation. J-Lens instead first applies a fixed, Jacobian-derived map that averages how activations affect downstream states across calibration contexts, rather than fitting a predictor to one task’s answers. Its intended target is verbalizable content, not accurate next-token prediction in every context (Gurnee et al., 2026).

Ouro gives this comparison two depth axes: physical layer and recurrent pass. The same layer stack is reused, so a readout that works at one pass need not work at another. The experiment asks where each method recovers a task-annotated intermediate concept more reliably. It does not assume that a recovered word was part of the model’s actual reasoning (Zhu et al., 2025).

My initial expectation was that J-Lens would provide a broadly better view of intermediate computation. That expectation did not survive the early-pass results under the final-target estimator. The positive finding came later: a specific region of the final pass where J-Lens outperformed raw readout. This paper separates

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the exploration that found that region from the new-question test that checked it, and reports the post-confirmation reviewer analyses separately from both.

The distinction matters beyond this comparison. Hidden-state readouts could complement observation of visible reasoning, but only after establishing what their scores measure. No monitoring system, deception detector, or chain-of-thought baseline is evaluated here.

Central claim. In one frozen recurrent model, one fixed Jacobian-lens estimator recovers annotated intermediate concepts from a fixed final-pass band far more often than the raw logit lens, and this transferred to 160 new questions without refitting or band selection. In the same physical layers of the three earlier passes the raw lens is better than this final-target estimator, and a lens targeting each pass’s own exit is better than both. Nothing here is evidence that the model uses the recovered content.

Organization. Section 2 places the study among readout and looped-model work. Section 3 defines the two readouts, the fitting recipe as executed, and the score. Section 4 separates the discovery work, which selected the band, from the frozen new-question test. Section 5 gives the confirmation result, then the post-confirmation reviewer analyses in their own subsection. Section 6 is the numerical verification, Section 7 the own-exit-target comparison, Section 8 the limits. Appendix A keeps the exploratory and Huginn material apart from the confirmation; Appendix B is the verification provenance and claim-status table; Appendix C the executed method settings.

2 Related work

Readouts of intermediate states. The logit lens applies a model’s output head to an intermediate residual state (nostalgebraist, 2020). The tuned lens replaces the head by an affine probe per block, trained to match the model’s output distribution (Belrose et al., 2023). The Jacobian lens (Gurnee et al., 2026) instead transports an intermediate state to the final layer with the average input–output Jacobian over a calibration corpus and decodes it with the model’s own unembedding; its authors report that on prompts with un verbalized intermediate computation the tuned lens tends to skip ahead to the output, and they find it less useful than either the logit lens or the Jacobian lens for inspecting internal computation. This paper compares only the two training-free readouts, the raw lens and J-Lens; Section 8 states why no tuned lens was fitted.

Looped and recurrent-depth models. Ouro applies one shared block stack over recurrent passes and trains it with weighted next-token losses at pass boundaries through one shared head (Zhu et al., 2025). Huginn trains a recurrent core between a prelude and a coda after sampled recurrence depths (Geiping et al., 2025); Lu et al. (2025) apply the logit lens to it and find its readability block-specific, the direct precedent for the raw-lens baseline here. Blayney et al. (2026) analyse the latent trajectories of looped models and report that many converge to layer-specific cyclic fixed points, with the recurrent block reproducing feedforward-like stages of inference; Popescu et al. (2026) study adaptive depth in looped transformers and compare learned halting gates with confidence-based readouts of the trajectory. Neither measures a fixed readout at fixed physical depth across passes, which is the quantity the same-band comparison here reports.

Prior readouts on Ouro. Kirin (2026) trains small supervised probes on frozen Ouro-RLTT trajectories to predict pre-answer success and candidate quality, and reports that recurrence moves where candidate quality can be read: early physical layers read weakly on the first pass and reach the late-layer basis by the third and fourth loops. Those probes are trained on task labels and read process quality; the readouts here are training-free and target annotated intermediate concepts on the base Ouro-2.6B checkpoint. The two results are complementary evidence that what can be read from Ouro at a given physical layer depends on the pass.

Why read latent passes. Chain-of-thought monitoring relies on reasoning that is externalized as text, and Korbak et al. (2025) argue that this monitorability is valuable and fragile. Models that compute in latent passes weaken that assumption, which is one reason to establish what a readout of those passes measures before anything is built on it. No monitor is built or evaluated here (Section 1).

3 Readouts and measurement

3.1 Model and fixed estimators

The model is **ByteDance/Ouro-2.6B**, revision 1ed04250da1a9936042725d302e81c8fa2ab5abd (not the Thinking or RLTT checkpoint), evaluated frozen in BF16 with four passes through 48 blocks, residual width 2,048 and a vocabulary of 49,152 tokens. I use one-based physical layers ℓ and pass numbers r ; the stored zero-based virtual index is $v = 48(r - 1) + (\ell - 1)$. Ouro re-normalizes the residual stream at the end of every pass before carrying it into the next pass and before the shared output head, so a recorded state at the last block of a pass is the pre-normalization block output.

For a captured residual state $h_{i,r,\ell}$ at the readout position of item i , let N be the model’s final RMS normalization and W_U its unembedding. The two vocabulary readouts are

$$z_{i,r,\ell}^{\text{raw}} = W_U N(h_{i,r,\ell}), \quad z_{i,r,\ell}^J = W_U N(B_{r,\ell} h_{i,r,\ell}),$$

where $B_{r,\ell}$ is the frozen J-Lens matrix for that virtual layer. Ranking the logits gives the token readout; no task classifier is trained on the confirmation questions. The main banks target the final residual state of pass 4 (virtual layer 191, before the final normalization). The original J-Lens study’s default Sonnet configuration targets the penultimate layer, so this is an adaptation, not an exact reproduction of that configuration (Gurnee et al., 2026).

Fitting recipe (from the executed records). Each bank averages, over 100 calibration paragraphs, the Jacobian estimator of the released implementation: for every source layer v and output dimension, a one-hot cotangent is placed at every valid target position of the target layer at once and back-propagated through the BF16 model with ordinary autograd, giving at source position p the sum over later positions $t' \geq p$ of $\partial h_{191}[t'] / \partial h_v[p]$; the mean over valid source positions (positions 16 to len−2 of a 128-token input, BOS included) is the paragraph’s matrix. Matrices are accumulated as an equal-paragraph FP32 sum, divided by 100 once and stored in FP16; at readout they are cast back to FP32. Calibration paragraphs for fit01–fit05 are independent random samples of 100 WikiText-103 training paragraphs (at least 600 characters; seeds 2026090701–05); the realized paragraph sets are pairwise disjoint and article overlap is 0–3 pairs per fit pair. **fit01** was preselected for confirmation before any new-item outcome; fit02 was retained as a secondary comparison. Fitting ran on an RTX 5090 with an 8-direction CUDA-graph engine whose output was verified bit for bit against the released fitter. Appendix C and the accompanying methods ledger give every setting with its source.

Two further comparisons examine estimator specification. The **target control** changes the derivative target from the final block (virtual 191) to the penultimate block (190) while retaining final normalization and unembedding. The **position controls** place the cotangent at one uniformly drawn valid position q per paragraph and keep all 2,048 derivative directions: *sampled-sum* sums the gradient over all valid source positions (its expectation over q equals the dense estimator), whereas *diagonal* keeps only the source position $p = q$. These are distinct estimator constructions, all fitted on fit01’s calibration paragraphs.

3.2 Prompt, readout position and what counts as recovery

Each question is one unfinished “Fact:” statement ending in a space; the intermediate concept is implied by the clue and never named, and the answer is the continuation. The model input is BOS plus the prompt tokens up to the boundary where the prompt’s tokenization stops agreeing with the tokenization of prompt-plus-answer (in 131 of 160 items the trailing-space token merges into the answer’s first token and is dropped); the readout position is the last input token, so every input token is causally visible to the scored state. There is no instruction, chat template, demonstration, prefilled response or padding; inputs are 13–35 tokens. Example (confirmation-001): “Fact: The planet whose sidereal rotation takes longer than its orbital period has an atmosphere dominated by” → readout at “by” (token 20), intermediate **Venus** (scored token 16293, “Venus”), answer “carbon dioxide”, never shown.

A concept is recovered when at least one accepted single-token form (the name with and without a leading space, capitalized or lower-cased, 1–4 accepted single-token forms per name) ranks among the first ten tokens of the **full vocabulary**. Each item’s controls are the other 79 intermediate names, equally weighted. Let $H_{i,r,\ell}^m(a)$ be the hit indicator for name a under method m , and a_i the intended name. With control set C_i , the fixed-layer excess score is

$$e_{i,r,\ell}^m = H_{i,r,\ell}^m(a_i) - \frac{1}{79} \sum_{c \in C_i} H_{i,r,\ell}^m(c).$$

The primary endpoint is the equal-weight item mean of the paired difference, averaged across the twelve fixed layers of the fourth pass:

$$\hat{\Delta} = \frac{1}{160} \sum_{i=1}^{160} \frac{1}{12} \sum_{\ell=26}^{37} (e_{i,4,\ell}^J - e_{i,4,\ell}^{\text{raw}}).$$

This averages fixed-layer measurements; it does **not** ask whether a concept appears anywhere in the band. The historical *any-layer* score separately gives each intended name and each control 48 opportunities per pass, then averages controls, labels within items, and items; it is reported separately.

All effects below are in percentage points. Recovery rates are averages over item–layer cells, not the fraction of questions the model answers correctly. Subtracting control recovery makes the comparison more informative, but does not turn it into a causal test or a calibrated measure of semantic correctness.

4 From discovery to a frozen new-question test

4.1 What the discovery work established

The discovery evaluation contained 90 eligible multihop items (100 labels) and 51 arithmetic items from the released lens-eval sets. Fixed-layer analysis showed that the early multihop loss was not merely an artifact of allowing the raw lens to recover a concept at a favourable late layer: the disadvantage remained in the middle of the stack. Arithmetic was mixed.

Pass 4 differed. Across five independent calibration fits, the mean excess advantage in layers 26–37 was **18.72 points** (between-fit SD **0.39**), positive in every fit, and every fit peaked at physical layer 32 (Figure 1). The final third (layers 33–48) averaged 6.13 points (SD 0.21). The gain predominantly reflected increased intended-concept recovery (18.91 own-name minus 0.19 control points).

That was evidence about calibration variability on the **same questions**, not an independent replication across evaluation populations. The band’s crossed fit/item pointwise interval was 10.96–26.99 points, but its simultaneous interval over the 56 prespecified metrics (−0.86 to 38.29) and its shared-concept interval (−7.49 to 44.92) contained zero, and no positive individual layer cleared the simultaneous shared-concept band. The band was therefore treated as a discovery to test, not a settled advantage.

4.2 Confirmation population and inference

The confirmation set contains **160 two-hop questions, 80 intermediate concepts, two questions per concept**, in ten domains of eight concepts (arts and music, astronomy, biology, chemistry, countries, environment and geology, historical people, landforms, SI units, technology). All 160 were eligible; none of the 80 intermediate concepts appears in the discovery set. The alias audit found one exact metadata collision — “N”, recorded as an alias of Newton, against the historical label “N” — which is not among the scored forms; no other alias collides. Construction and annotation were blind to method outcomes; factual hops, novelty and control hygiene were reviewed independently before the local freeze.

The relation mixture differs from discovery: 96 questions request relation types absent from the discovery inventory, 45 use familiar requested relations, 19 recombine familiar relations. This is a transfer test, not a

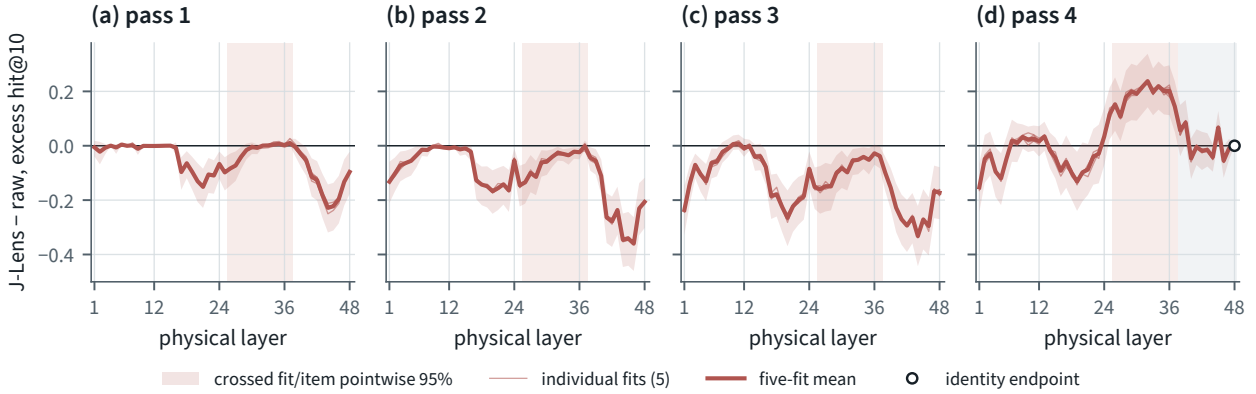


Figure 1: Discovery population: fit01–fit05 minus raw, fixed-layer excess hit@10 for the 90 eligible multihop items, all four passes. Thin lines are the five fits, the bold line their mean, shading the crossed fit/item pointwise 95% interval (not simultaneous). Pale red: layers 26–37, selected here and later frozen; grey: final third. The open point at layer 48 is the identity endpoint.

difficulty-matched replication. Three component facts overlap known calibration text (items 076, 130, 151). No claim is made that the facts were absent from pretraining.

Fit01, the pass-4 band, scoring rules, population and analysis were fixed before any new-item outcome; the stopping rule was one evaluation of that population. “Prospective” refers to that recorded local freeze (9 September 2026, 22:40:58 UTC); a public preregistration does not exist. Two runtime corrections frozen before scoring and the later numerical checks are described in Appendix B.

The primary interval uses **20,000 bootstrap resamples of whole dependency groups** (seed 2026090901), with the 2.5th and 97.5th percentiles as limits; resampled groups contribute all their items and the ratio of sums preserves equal item weights. Twenty prespecified secondary contrasts form a separate simultaneous 95% family (bootstrap max- t). Component intervals are descriptive.

Dependency groups. Items are joined into a group when they share an intermediate (or its aliases), an underlying fact, a concrete entity-substitution template (the same typed clue $\rightarrow$ entity $\rightarrow$ requested-slot construction, such as element clue $\rightarrow$ element $\rightarrow$ atomic number), or a normalized final answer or alias code. The 28 connected components have sizes 42, 24, 16, 10, 10, 6, four of 4 and eighteen of 2; the weight-based effective group count is $160^2 / \sum_g n_g^2 = 8.63$. The largest group joins the 16 chemistry items (element $\rightarrow$ atomic-number and element $\rightarrow$ symbol templates) and the 16 SI-unit items (unit $\rightarrow$ symbol and unit $\rightarrow$ quantity templates) through tungsten’s and Watt’s shared answer “W”, with three small clusters (Alps/Andes/Borneo; nucleus; ozone) attached by the small-integer answers “2” and “3”. Leaving out any single group keeps the estimate between 19.73 and 24.87 points (low end without the 16 country items, high end without the 42-item group). The effective count is a weight summary, not a literal number of independent observations, and the grouping does not certify the absence of other dependence.

5 Results on new questions

5.1 The fixed final-pass advantage transfers

Result 1: the frozen pass-4 band transfers to new questions. In pass 4, layers 26–37, fit01 exceeds the raw lens by **23.19 points** in excess hit@10, 95% dependency-group interval **16.29–32.90** (Table 1, Figure 2). The gain comes from intended-concept recovery (37.92% against 14.43%); control recovery rises slightly (+0.30 points), reducing rather than creating the advantage. Item resampling gives 17.97–28.71; the secondary fit02 gives 22.56 (9.52–35.59) and fit01 – fit02 is 0.63 (−0.73 to +1.99).

Table 1: Confirmation results for the frozen pass-4 band. Component intervals are descriptive; the final row is the primary inference. Rounded entries need not subtract exactly.

Quantity	Estimate	95% group-bootstrap interval
J-Lens intended recovery	37.92%	30.44–46.56%
Raw intended recovery	14.43%	10.45–17.87%
J-Lens control recovery	0.34%	0.18–0.51%
Raw control recovery	0.04%	0.02–0.08%
Paired excess difference	+23.19 points	+16.29 to +32.90 points

The intended-recovery difference is **23.49 points** (16.62–33.16, descriptive). The two fixed estimators perform similarly here; this is not a formal equivalence result. Discovery and confirmation results are not pooled.

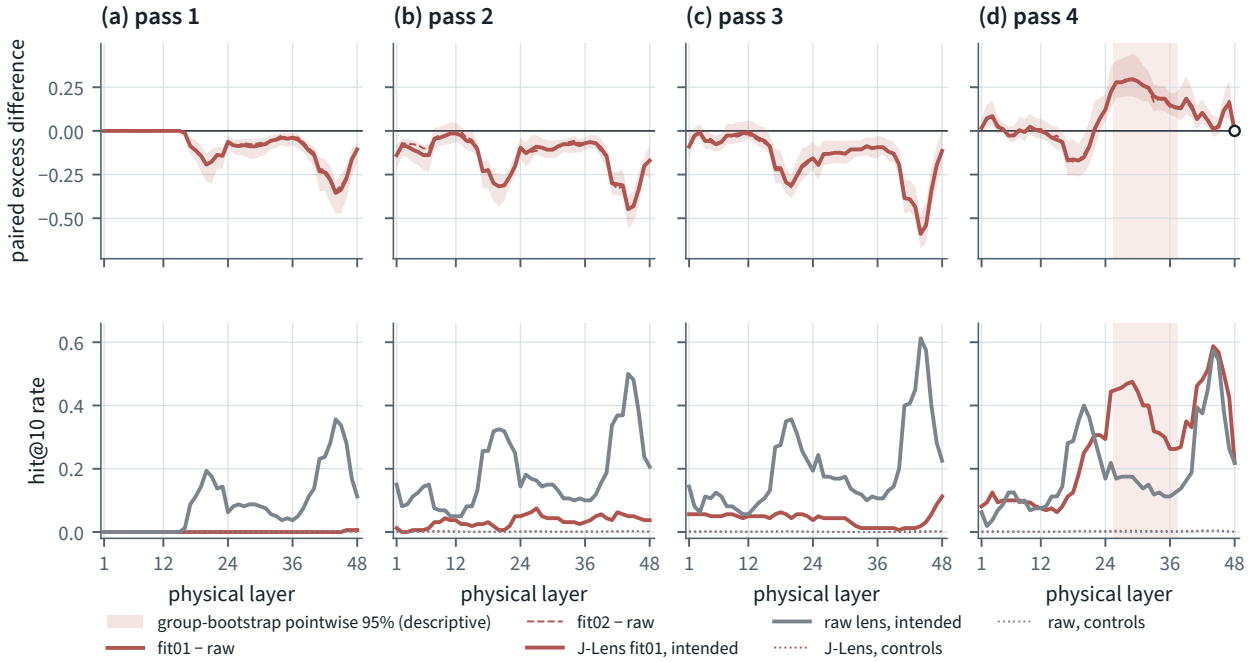


Figure 2: Confirmation population, fixed-layer curves for all four passes. Top: fit01 – raw (solid) and fit02 – raw (dashed) paired excess difference per layer, with the descriptive group-bootstrap pointwise 95% band for fit01. Bottom: intended recovery for fit01 (red) and the raw lens (grey), with control recovery dotted. Pale red: the frozen primary band. Only the band mean carries the prospective inference.

5.2 The early-pass deficit transfers, including in the same physical band

All six prespecified early-pass comparisons remain negative on the new population (Table 2).

These summaries alone do not show what happens in the specific band that carries the pass-4 gain. As a **post-confirmation reviewer analysis**, the same physical layers 26–37 were evaluated in every pass with the frozen estimator, controls and groups (Table 3, Figure 3).

Result 2: the same band is negative in every earlier pass. Layers 26–37 show a deficit in each early pass whose simultaneous interval (a four-contrast post-confirmation family, Appendix B) excludes zero: -6.50 , -8.64 and -11.14 points in passes 1–3. In pass 1 fit01 recovers no intended label in the band at all. The pattern is pass-dependent at fixed physical depth, not an artifact of averaging over 48 layers.

Table 2: Fit01 minus raw excess hit@10, in percentage points. Brackets are simultaneous 95% intervals from the 20-contrast secondary family.

Pass	Mean over fixed layers 1–48	Any-layer recovery
1	−8.83 [−13.64, −4.01]	−44.41 [−59.03, −29.78]
2	−14.86 [−19.54, −10.18]	−57.37 [−72.01, −42.72]
3	−15.59 [−22.40, −8.77]	−61.08 [−79.59, −42.58]

Table 3: The same physical band, layers 26–37, in every pass. Control recovery is below 0.35% in every cell.

Pass	J-Lens intended	Raw intended	Paired excess difference	Interval	Status
1	0.00%	6.46%	−6.50	[−11.91, −1.08]	post-confirmation, simultaneous
2	4.22%	12.71%	−8.64	[−12.43, −4.85]	post-confirmation, simultaneous
3	2.76%	13.91%	−11.14	[−16.76, −5.52]	post-confirmation, simultaneous
4	37.92%	14.43%	+23.19	[+16.29, +32.90]	prospective primary

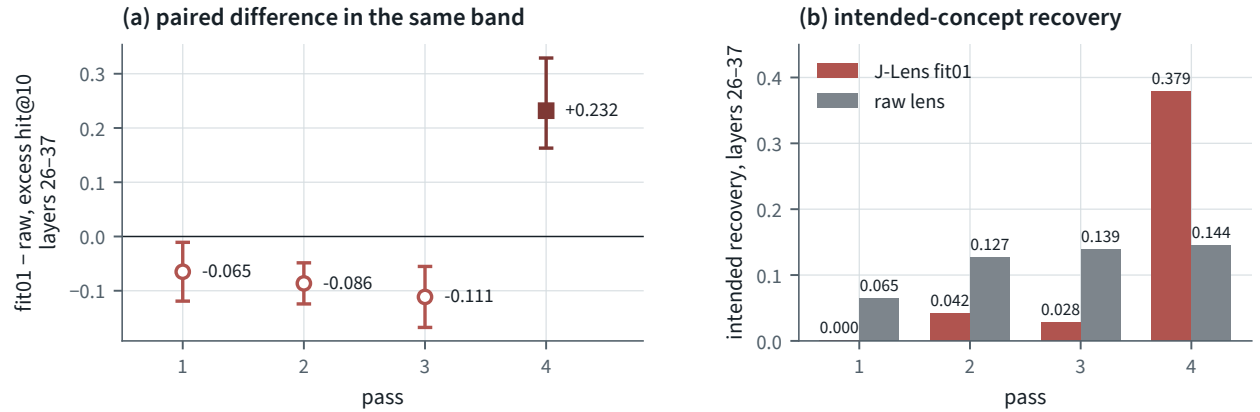


Figure 3: Layers 26–37 in every pass on the confirmation population. (a) Paired fit01 – raw excess difference; passes 1–3 (open circles) carry the post-confirmation simultaneous 95% interval of family N1, pass 4 (filled square) the original prospective percentile interval. (b) Intended recovery by method.

5.3 The estimator specification matters

The control comparisons transfer to the new questions (Table 4, Figure 4).

Table 4: Estimator controls in percentage points; matched layer support within each control family; same 20-contrast family as Table 2.

Contrast in pass 4, layers 26–37	Difference	Simultaneous 95% interval
Main fit01 minus penultimate target	+14.38	[+7.70, +21.07]
Sampled-sum minus diagonal	+15.77	[+5.80, +25.75]
Penultimate target minus raw	+8.80	[+0.76, +16.85]
Sampled-sum minus raw	+21.72	[+9.16, +34.29]
Diagonal minus raw	+5.95	[−0.34, +12.24]

Result 3: the late gain depends on the derivative target and the position aggregation. Changing the derivative target by one block reduces the advantage substantially; restricting the position reduction to the matching source position does so as well. Neither comparison isolates how Ouro reasons: it shows that the measured recovery depends on the lens’s construction. The diagonal-only estimate is smaller and its support is borderline (lower bound within 0.4 points of zero on every numerical path, excluding zero only under regenerated states). The direct sampled-sum-minus-diagonal comparison is the clearer result.

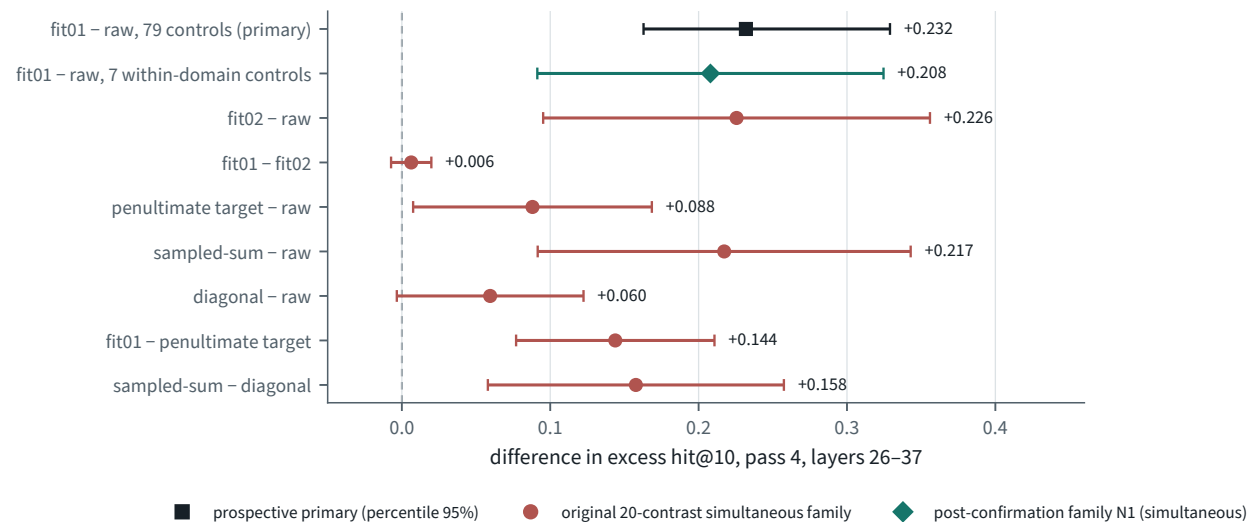


Figure 4: Pass-4 band contrasts; the dashed vertical line marks zero difference. Black square: the prospective primary (percentile interval). Red circles: contrasts of the original 20-member simultaneous family. Teal diamond: the post-confirmation within-domain-control contrast (Section 5.4, family N1).

5.4 Post-confirmation reviewer checks

Three further analyses were specified on the already-used confirmation data after the confirmation; they are reported separately and do not alter the primary endpoint.

Within-domain controls. Replacing each item’s 79 controls by the seven other names in its benchmark domain gives a paired excess difference of **20.80 points** (simultaneous 95% interval 9.13–32.46). J-Lens recovers same-domain names at 3.07% against 0.34% for arbitrary controls (raw: 0.38% against 0.04%). Using within-domain controls therefore reduces the advantage by 2.39 points, from 23.19 to 20.80: the lens recovers more competing names from the same domain, but the within-domain-adjusted advantage remains positive. This measures the sensitivity of the excess score to the control set; it does not identify

a unique fraction of the signal as type-level information. A subset inequality derived from the published values (an item’s within-domain excess difference is at least its intended difference minus 79/7 times its J-Lens all-control rate) held for every item; averaging its itemwise lower bounds gives +19.61 points, below the observed aggregate +20.80-point advantage. Four of the ten domains contain a single semantic type (elements, countries, SI units, people); the others mix types (planets and moons; rivers, ranges, a desert and an island; molecules, a cell type and an organelle), so this is a within-domain control, not a same-type control. The gain is heterogeneous across domains: from about +51 points for countries and +40 for environment and geology to about zero for SI units, where neither method recovers unit names in this band, and to −1 for astronomy (+5 with the 79 controls), where the lens often recovers other planets and moons. Domains are not independence units.

Tokenized-input overlap. For every item, every scored token of the intended label and of all 79 controls was compared with the exact model input. No intended-label token appears in any input. Six control tokens do, in six items: “Voyager” in three astronomy prompts, “Chile” in an Andes prompt, “Egypt” in a Nile prompt, and the lower-case form “io” as a token inside “Fidelio”. Dropping those controls for the affected items (both methods; 78 controls for six items) gives 23.19 points (16.25–32.72); dropping the six items gives 22.90 points (15.92–33.15). Lexical word-level mentions without scored-token overlap occur once more (“newton” in a pascal item). Zero token overlap does not establish absence of semantic clues or of the facts from pretraining; the prompts are designed to imply the intermediate.

Grouping. The rule, memberships and the composition of the 42-item group are given in Section 4.2 and the reviewer ledger; no alternative grouping was constructed.

6 Numerical verification

Audit note: post-confirmation, not prospectively registered. The confirmation run used BF16 model execution on an RTX 5090. A development check initially failed because applying the output head to one residual row did not reproduce logits computed in a multi-row call. The verification below measured whether this affected the scored comparisons rather than assuming a small logit discrepancy was harmless.

On a local RTX 5070 Ti with the confirmation’s torch build and precision settings, the retained confirmation states, verified banks and frozen scoring code reproduced the saved arrays bit for bit across **160 items, six arms and all scored columns**, including ranks, top-ten token IDs, saved exit logits and sampled transported vectors. Calls with 160, 190, 191 and 192 rows matched the executed layout exactly; single-row calls changed 1.5–1.6 million of about 1.5 billion logits per arm by at most 0.125 (0.25 for raw) but no endpoint quantity.

Table 5: Numerical sensitivity of the primary endpoint. The FP64 row applies a reference readout computation to the stored tensors, not an FP64 model. Alternative packing was measured on the local GPU, not in a fresh RTX 5090 run.

Computation	Primary difference	95% group interval	Change from accepted
Accepted BF16 path	+23.19	[+16.29, +32.90]	—
Alternative row packing	+23.19	unchanged	0.00
FP64 reference computation	+22.93	[+16.17, +32.52]	−0.26
Locally regenerated states	+23.61	[+16.52, +33.77]	+0.42

CPU and GPU FP64 computations gave identical ranks, hits and top-ten sets (values differing by at most 9×10^{-14}). Resolving every exactly tied logit adversarially bounds the accepted estimate between **22.60 and 23.35 points**; moving every logit by up to 1/32 or 1/8 in the adversarial direction gives 22.01–24.41 and 18.30–27.26. Regenerating states from the prompts on the local GPU was not bit-identical to the RTX 5090 forward: 37 of 160 state records matched exactly and 154 greedy continuations matched, yet replaying every arm on those states moved the endpoint by only 0.42 points. All six early-pass contrasts and the direct

estimator-control contrasts keep their signs and zero-exclusion on every path. Bit-level state regeneration on another RTX 5090 was not tested.

A separately implemented checker (a separate coding agent working from a specification, without producer code) reconstructed the saved analysis to within 9×10^{-16} and detected all 13 planted errors in isolated fixtures. “Independent” here means a separate implementation, not replication by another laboratory.

7 Own-exit targets: the early deficit belongs to the target

The main bank targets the final pass, so early-pass states are far from its target: a pass-1 state at layer 30 is 162 blocks and three pass-boundary normalizations away from virtual layer 191. One explanation of the early deficit is therefore transport distance rather than a property of early-pass representations. The test evaluates the same stored early-pass confirmation states with the raw lens, a J-Lens targeting the current pass’s own exit, and the final-target J-Lens, in layers 26–37 of passes 1–3.

Banks. The only own-exit banks are the initial-study family (5 September 2026): banks targeting virtual 47, 95, 143 and 191, fitted on the first 100 paragraphs of the historical calibration file (12 articles) with one shared recipe that differs only in the target, so within this family a difference between banks is a difference of target. Their merged files had been purged. On 12 September the thirteen surviving shards were retrieved from the author’s private repository, each verified against its recorded hash, and merged with the frozen count-weighted procedure; the merge of the final-target bank equals the surviving pod merge entry for entry. The family’s calibration (12-article prefix, 32 directions per backward pass) differs from fit01’s, so own-exit – fit01 is reported only descriptively. The inferential contrasts, fixed in a dated plan before the banks were applied, are own-exit – final-target (exit 3) and own-exit – raw in the band for passes 1–3, one simultaneous family.

Result 4: own-exit targets reverse the early deficit on the confirmation population. Applied to the accepted confirmation states with the frozen scorer, controls and groups, the own-exit lens recovers the intended concept in the band in 23.1%, 31.3% and 37.4% of item–layer cells in passes 1–3, against 6.5%, 12.7% and 13.9% for the raw lens and 0.6%, 4.7% and 2.9% for the final-target bank of the same family; control recovery stays below 0.4%. Own exit minus final target: **+22.60, +26.50 and +34.17 points**; own exit minus raw: **+16.41, +18.32 and +23.18 points** (Table 6, Figure 5). All six simultaneous 95% intervals exclude zero (one max-t family, 20,000 whole-group draws, quantile 2.699). In pass 4 the same family’s final-target bank reproduces the confirmed gain: exit 3 minus raw +22.39 points (16.07–31.68, descriptive) against fit01’s +23.19.

Table 6: Own-exit versus final-target and raw readouts in layers 26–37 on the confirmation population, initial-study family. Brackets: simultaneous 95% intervals of the six-contrast post-confirmation family. Intended recovery per item–layer cell; control recovery is below 0.4% in every cell.

Pass	Own exit – final target	Own exit – raw	Own-exit intended	Final-target intended	Raw intended
1	+22.60 [+11.61, +33.60]	+16.41 [+8.92, +23.91]	23.12%	0.57%	6.46%
2	+26.50 [+18.37, +34.63]	+18.32 [+10.53, +26.11]	31.30%	4.69%	12.71%
3	+34.17 [+23.39, +44.94]	+23.18 [+11.49, +34.87]	37.40%	2.92%	13.91%

Interpretation. Within the rules fixed before these numbers were computed, this is the case in which the own-exit target beats both alternatives on the confirmation population: a target-dependent early-pass advantage, consistent with a transport explanation, now with simultaneous intervals rather than the descriptive discovery-population result that preceded it (Appendix A). It is not proof that averaging destroyed information, and it is a post-confirmation analysis on the population already used for the primary endpoint. The two estimator families differ in calibration; the agreement of their final-target banks in pass 4 (−0.80

points, -1.71 to $+0.30$) limits how much of the early-pass difference calibration could explain, but does not remove that caveat. Two existing qualifications remain: shortening the final target by one block (the penultimate control) reduced the late gain, so transport distance alone is not established, and raw pass counts are not comparable transport lengths across architectures. The observations are jointly consistent with a narrower hypothesis, not tested here: that the estimator reads well when its derivative target is an exit state — the tensor the shared head actually reads after the pass-end normalization — and degrades with distance from the nearest such target, so that moving the target off an exit by one block can cost more than moving the source a full pass away from it. That hypothesis predicts that a lens targeting the penultimate block of an early pass would read that pass worse than one targeting the pass’s own exit; the retained banks do not include such a control. The historical Huginn pilot (Appendix A) did not identify a cause or rule out supervision-based explanations.

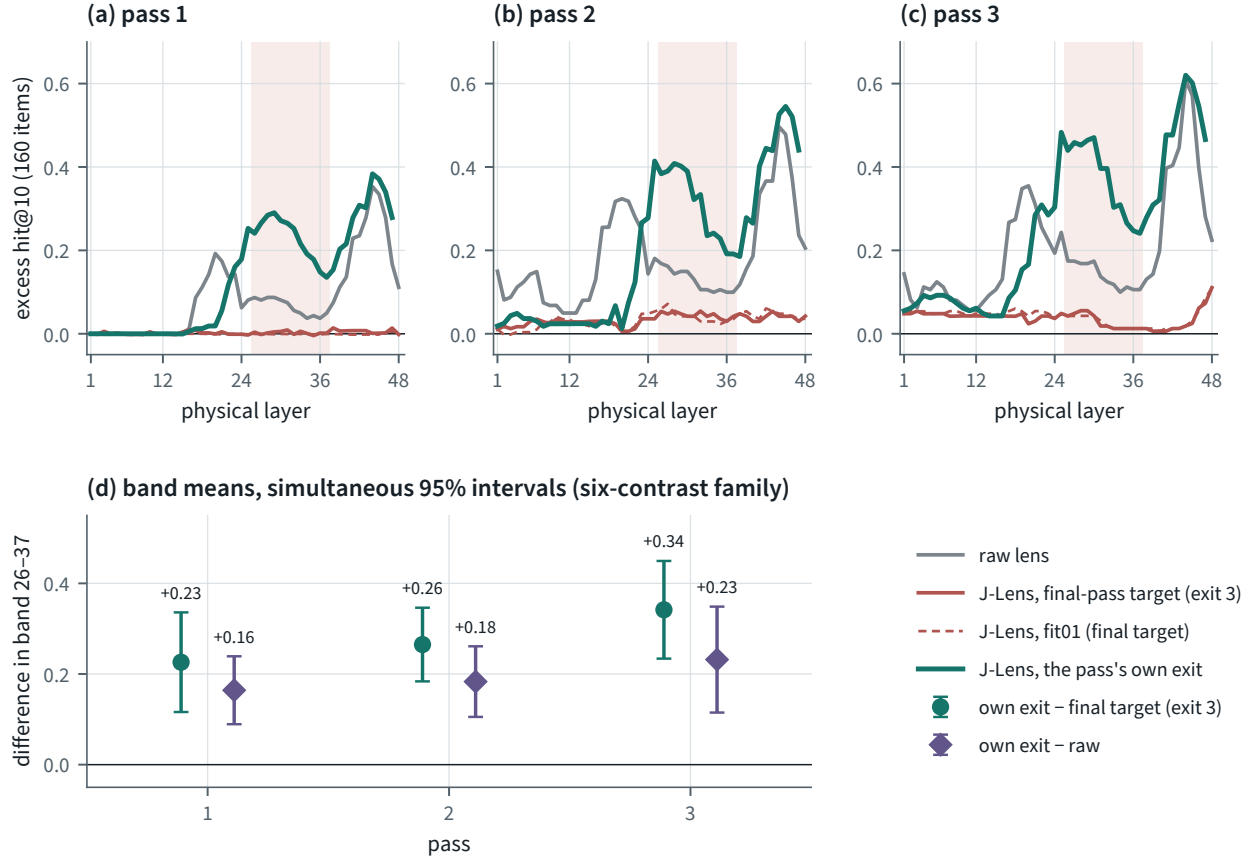


Figure 5: Initial-study family on the confirmation population, fixed layers, passes 1–3: raw lens (grey), final-target J-Lens (exit 3, red), fit01 (dashed red) and the pass’s own-exit J-Lens (teal). (d) The six inferential contrasts, band means with simultaneous 95% intervals. Post-confirmation analysis; the plan was fixed before the retrieved banks were applied.

8 Interpretation and limits

The supported result is a **localized improvement in reading annotated intermediate concepts from fixed Ouro states**: a specific final-pass band, one frozen estimator, transferred to new questions and concepts without refitting or band selection. The same band in the earlier passes shows the opposite sign for the final-target estimator, so that estimator is not uniformly better as computation proceeds; with the target moved to each pass’s own exit the sign reverses, so the deficit is a property of the target. The source

of the measured gain is higher intended-concept recovery; the excess score is sensitive to the control set, with within-domain controls reducing the advantage by 2.39 points while leaving it positive.

One possible explanation for the strong raw baseline is Ouro’s training objective, which applies weighted next-token losses at recurrent pass boundaries through a shared output head (Zhu et al., 2025). This does not supervise every physical block, and the present experiments do not isolate the effect of that objective. The target-dependence on the confirmation population isolates target choice within one estimator family, but architecture, the training objective and executed depth remain entangled, and the Huginn pilot’s evidence is limited by its incomplete artifact chain.

Recovery is not use. In exploratory checks a lens recovered a labelled intermediate on questions the model answered incorrectly, and native exits can agree while their readouts disagree sharply (Appendix A).

No tuned lens. The comparison includes the two training-free readouts and no learned alternative. A tuned lens fits a per-layer map to the model’s output distribution (Belrose et al., 2023); it asks whether a trained probe can predict the output from an intermediate state, whereas the question here is whether a fixed Jacobian transport recovers intermediate content that the head the model was trained with misses. The two questions have different reference points, and the J-Lens authors report that the tuned lens tends to skip ahead to the output on prompts with un verbalized intermediates (Gurnee et al., 2026). A tuned-lens arm on the frozen band, twelve layers in each of four passes, would show whether a data-trained map recovers the same intermediates; it has not been run, and it is the first limitation of the present comparison.

The main population limit is the small number of unequal dependency groups. The interval accounts for the declared grouping, not every possible shared fact, relation or template; the effective count of 8.63 warns against reading 160 related questions as 160 demonstrations. The relation mixture is not difficulty-matched, labels are restricted to accepted single-token forms, the domain heterogeneity is large, and the evidence concerns one checkpoint and two retained calibration fits. The study does not establish performance on arbitrary concepts, other models, or monitoring under adversarial pressure.

8.1 Reproducibility and artifact availability

The confirmation banks, states, scores, frozen inputs and code permit exact table reconstruction, rescoring from banks and states, and forward replay (not bit-identical off the RTX 5090). Complete refitting of the original estimators is not available: an earlier retrieval failure and a later storage purge destroyed unique historical artifacts, including all four Ouro FP32 fitting checkpoints, the historical discovery-evaluation state cache, the truncated Huginn checkpoint and the merged files of the initial-study local-exit banks (43 purged local paths, 33 distinct contents, with no copy found and no verified derivation). Of a further 19 purged paths (15 distinct contents) whose records pointed to previously uploaded copies in a private remote repository, the thirteen own-exit bank shards were retrieved from that repository on 12 September 2026 and verified against their recorded hashes; the rest remain unverified. Knowing the recipe does not restore those checkpoints, and the Huginn pilot is retained only at the saved-output level. On 11 September 2026 a copy-only archive of every surviving file (16,200 files, 52.15 GB, every byte re-hashed on the destination) was written to one external SSD and restoration-tested from that SSD alone with the laptop’s paths hidden and no network: the frozen code matched its freeze, all four bank files loaded, the frozen analysis reproduced the primary and all twenty secondary contrasts exactly, and raw and fit01 rescoring from the archived states and weights reproduced the saved ranks bit for bit. Until a second destination exists it is a single-copy archive, not a redundant backup. The frozen records, code, compact results, figures and this manuscript are included in the supplementary material (files of 50 MB or more, including the banks and state caches, are listed there with hashes but not stored). No DOI exists at the time of writing.

9 Conclusion

The final-target J-Lens underperforms raw readout in Ouro’s first three passes, including in the very layers where it outperforms raw in the fourth, and recovers more annotated intermediate content in a fixed region of the final pass. That advantage transferred to new questions and concepts and survives the measured numerical variations and control redefinitions. Its size depends strongly on the derivative target and the

position aggregation, and on the confirmation population a lens targeting each pass’s own exit reads early passes far better than both the final-target lens and the raw lens, so the early deficit belongs to the estimator’s target rather than to the passes. The finding is about where a particular readout works, not about general superiority, successful reasoning or causal control.

Broader Impact Statement

This paper evaluates a readout method on one frozen open-weights model. Hidden-state readouts could complement observation of visible reasoning, but only after establishing what their scores measure. No monitoring system, deception detector, or chain-of-thought baseline is evaluated here, and the study does not establish performance on arbitrary concepts, other models, or monitoring under adversarial pressure.

References

- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2023. URL <https://arxiv.org/abs/2303.08112>.
- Hugh Blayney, Álvaro Arroyo, Johan Obando-Ceron, Pablo Samuel Castro, Aaron Courville, Michael M. Bronstein, and Xiaowen Dong. A mechanistic analysis of looped reasoning language models, 2026. URL <https://arxiv.org/abs/2604.11791>.
- Jonas Geiping et al. Scaling up test-time compute with latent reasoning: A recurrent depth approach, 2025. URL <https://arxiv.org/abs/2502.05171>. Sections 3.1–3.3 and 6.1 for the Huginn architecture, training and early exits.
- Wes Gurnee, Nicholas Sofroniew, Adam Pearce, et al. Verbalizable representations form a global workspace in language models. Transformer Circuits, Anthropic, July 2026. URL <https://transformer-circuits.pub/2026/workspace/>. Methods; Quantitative Methodological Comparisons; Methodological Details and Ablations.
- Jan Kirin. Operational proto-introspection in looped language models: Process-quality taps, executable branching, and the readout–control boundary, 2026. URL <https://arxiv.org/abs/2607.18553>.
- Tomek Korbak, Mikita Balesni, Elizabeth Barnes, et al. Chain of thought monitorability: A new and fragile opportunity for AI safety, 2025. URL <https://arxiv.org/abs/2507.11473>.
- Wenquan Lu et al. Latent chain-of-thought? decoding the depth-recurrent transformer, 2025. URL <https://arxiv.org/abs/2507.02199>. Section 3.2 for block-specific logit-lens readability in Huginn.
- nostalgebraist. Interpreting GPT: the logit lens. LessWrong, August 2020. URL <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>.
- Andrei Cristian Popescu, Haitz Sáez de Ocáriz Borde, and Pietro Liò. Adaptive depth in looped transformers: Diagnosing learned halting gates and trajectory readouts, 2026. URL <https://arxiv.org/abs/2607.20519>.
- Rui-Jie Zhu, Zixuan Wang, Kai Hua, et al. Scaling latent reasoning via looped language models, 2025. URL <https://arxiv.org/abs/2510.25741>. Sections 3.1–3.3 for the architecture and training objective.

A Exploratory findings kept separate from confirmation

A.1 Exit disagreement

In the 7 September analysis of 148 prompts, native pass-3 and final predictions agreed **85.1%** of the time (item-bootstrap interval 79.1–90.5%). Two J-Lens variants targeting the current and final exits agreed only **1.45%** (0.58–2.51%) averaged over physical layers 41–47; layer 48 was excluded because some agreement

there follows by construction. One token earlier, native agreement rose to 96.6% while lens agreement remained 0.39%. These are different comparisons, so the lens gap cannot be read as the amount of model revision, and poor output prediction does not by itself refute an intermediate-recovery method (Gurnee et al., 2026).

A.2 Passing continuations and arithmetic

The early multihop deficit remained when the discovery items were separated by the historical continuation-pass criterion. That criterion was imperfect: two numerical false positives reduced the original 72 passes out of 148 to 70, while additionally accepting the numeric equivalent of one word-form target gave 71. Four-token continuations and lexical matching do not establish semantic correctness; the subgroup observations are exploratory.

A separate supervised arithmetic probe scored 59.4% against the raw lens’s 76.4% in pass 1, but exceeded it in pass 3 (35.6% versus 17.9%). The classifiers trained on only 36 independent operand pairs; a bounded training-size comparison (+3.9 points, interval -2.9 to $+10.9$) was inconclusive. This used a different task family and a candidate-set metric, with incomplete provenance for the older J-Lens comparator; it is not pooled with the concept-recovery results.

A.3 Own-exit targets on the discovery population

Before the confirmation-population test of Section 7, the initial-study banks had been applied to the 148 discovery items in one evaluation whose rank arrays survive. On the 90 eligible multihop items, in the fixed band 26–37, the lens targeting each pass’s own exit recovers the intended concept far more often than both alternatives: intended recovery 15.4%, 16.2% and 25.2% in passes 1–3, against 2.1%, 4.3% and 10.1% for the raw lens and 0.3%, 0.4% and 2.1% for the final-target lens. Own exit minus final target: +14.6, +16.9 and +23.0 points (shared-concept-cluster percentile intervals 9.1–20.9, 11.5–22.8, 16.0–30.6); own exit minus raw: +13.0, +11.3 and +14.9 (8.1–19.0, 6.6–16.6, 7.9–22.4). Control recovery stays below 2.3%. These intervals are descriptive, the population is the one on which the band was selected, and the result is superseded by Section 7.

A.4 Huginn pilot

The pilot used one Huginn estimator calibrated on the same 100 texts as the preselected Ouro fit, eight recurrent passes and two evaluation initializations. Joint tokenizer eligibility retained 88 multihop items (98 labels) and 51 arithmetic items. J-Lens-minus-raw estimates were negative under all six prespecified summaries in both initializations; the cross-model differences of differences had mixed signs and simultaneous intervals containing zero. The prediction of a larger relative J-Lens advantage in Huginn was not supported. This does not isolate supervision: Huginn’s coda is trained after sampled recurrence depths (Geiping et al., 2025), so it is not an unsupervised-intermediate control, and its raw-readout weakness is block-specific (Lu et al., 2025). The fitted Huginn bank was never retrieved and its partial checkpoint was later lost; the saved statistics are historical evidence, not a verified cross-model replication.

B Verification provenance and claim status

B.1 Verification and analysis provenance

The original confirmation freeze is unchanged. Two separately recorded corrections preceded new-item scoring: `precision_v1` (10 September, 10:44:52 UTC) removed redundant TF32 setters, and `native_check_v1` (22:56:18 UTC) changed the development comparison from a single residual row to a whole-sequence read-out and resized run budgets. The checker’s accepted run-specification pin was updated after outcomes were known. The post-confirmation verification reviewed these changes and found the scientific input, estimator, endpoint and analysis unchanged.

The numerical results use the accepted BF16 endpoint as primary; FP64, alternate packing, tie handling and regenerated states are sensitivity checks. For diagonal minus raw, the accepted estimate is +5.95 points with simultaneous interval -0.34 to $+12.24$; FP64 gives -0.40 as lower bound and regenerated states $+0.04$, which is why its support is described as borderline.

Post-confirmation reviewer family (Sections 5.2 and 5.4). Exactly four new inferential contrasts were fixed in a dated plan before computation: the within-domain paired excess difference in the pass-4 band and the same-band paired differences in passes 1–3. They share one centered bootstrap max- t 95% family (20,000 whole-group draws, seed 2026091101, quantile 2.639). Every other new quantity is descriptive; the overlap sensitivities and the discovery-population own-exit summary (Appendix A) are descriptive.

Own-exit family (Section 7). Six further contrasts were fixed in a dated plan before the retrieved banks were applied to the confirmation states: own-exit – final-target (exit 3) and own-exit – raw in the band for passes 1–3. They share one centered bootstrap max- t 95% family (20,000 whole-group draws, seed 2026091201, quantile 2.699). The initial-study final-target bank’s pass-4 comparisons and own-exit – fit01 are descriptive. Before the banks were applied, the frozen scorer reproduced the accepted raw and fit01 readouts from the same states bit for bit.

Records. Confirmation run `ouro_confirmation_20260911_fixed160_native1`; frozen plan, benchmark and freeze receipt under `research/confirmation_2026-09-09/`; verification under `research/verification_2026-09-11/` (`FINAL_VERIFICATION.md`, `REPORT_confirmation_verified.md`, `RUN_SPECIFICATION.json`); this round’s reviewer checks, methods ledger, numerical source ledger and archive receipt under `research/closeout_2026-09-11/`; the own-exit analysis (plan, retrieval receipt, merge record, readouts, results) under `research/closeout_2026-09-11/local_exit/confirmation_2026-09-12/`. These are project provenance identifiers, not public links.

B.2 Claim status

Table 7: Claim status after the verification and reviewer rounds. Established = supported by the frozen or replayed evidence at the stated scope; Bounded screen = supported with a quantified limit; Diagnostic = descriptive observation, no inference; Unresolved = open; Retracted = an earlier statement withdrawn.

Claim	Status	Basis
Fit01’s pass-4 band advantage transfers to 160 new questions	ESTABLISHED	Prospective primary +23.19 [16.29, 32.90]; replay and FP64 paths leave it intact
The early-pass deficit transfers, including in the same band	ESTABLISHED	Six prespecified contrasts and the post-confirmation same-band family all exclude zero
The late gain survives within-domain controls	BOUNDED SCREEN	Within-domain controls reduce the advantage by 2.39 points (23.19 to 20.80) and leave it positive; heterogeneity across domains is large; no fraction of the signal is identified as type-level
Target choice and position aggregation change the late gain	ESTABLISHED	fit01 – penultimate and sampled-sum – diagonal exclude zero on every path
Diagonal fitting has no effect	UNRESOLVED	Diagonal – raw is borderline: lower bound within 0.4 points of zero
Own-exit targets read early passes better than the final target and raw	ESTABLISHED	Six-contrast post-confirmation family on the confirmation population, all exclude zero; one estimator family
Ouro’s shared-head supervision causes the strong raw baseline	UNRESOLVED	No intervention; Huginn pilot uninformative and estimator lost
Recovered concepts are causally used	UNRESOLVED	No causal test; readable intermediates co-occur with wrong answers
All historical estimator binaries were preserved	RETRACTED	Four FP32 checkpoints and the Huginn estimator are lost; the own-exit bank shards were retrieved on 12 September

C Method details in brief

C.1 Executed settings, with their record sources in the methods ledger

- **Tokenization and input.** BOS + prompt tokens (`truncation=True`, `max_length=511`, never triggered); readout context = common prefix of `encode(prompt)` and `encode(prompt+answer)`; position `-1`; batch size 1; SDPA attention; `use_cache=False`; BF16 weights.
- **Hooks and normalization.** Forward hook on each shared physical block filtered by the recurrent step; recorded tensor = post-residual block output, before the pass-end RMS normalization; states stored as FP32 copies of BF16 values.
- **Readout.** Jh in FP32 (raw: h), cast to BF16, RMS-normalize (FP32 internally, $\epsilon = 10^{-6}$), BF16 head, one call per item over all scored columns; descending argsort over 49,152 tokens; rank = min over accepted forms; hit = rank < 10.
- **Calibration and fitting.** 100 WikiText-103 paragraphs per fit (fit01–05 seeds 2026090701–05; paragraph-disjoint; 100/99/99/100/100 distinct articles), inputs of 128 tokens including BOS (six shorter), valid positions 16 to len–2, cotangent summed over valid target positions and gradient averaged over valid source positions, FP32 sum over paragraphs divided by 100, FP16 storage, 8 output directions per backward pass with CUDA-graph replay (bit-parity verified), RTX 5090.

- **Controls.** Penultimate: target virtual 190, sources 0–189. Positions: one uniform q per paragraph (seed 2026090801), sampled-sum = $\sum_p \partial h_{191}[q]/\partial h_v[p]$ (no divisor), diagonal = $\partial h_{191}[q]/\partial h_v[q]$, from one backward stream. Matched supports for contrasts: band 26–37 for all arms; final-third intersections 176–189 (target control) and 176–190 (position controls).
- **Banks.** fit01 90f01f6a..., fit02 101f31db... (byte-exact reconstruction from its intact checkpoint), penultimate dc6354df..., positions b8e8b7d2... (both position arms); four files, five bank arms, six evaluated arms.
- **Initial-study family (Section 7).** Same estimator with the first 100 historical paragraphs (12 articles), 32 directions per backward pass, B300 pod; targets 47, 95, 143, 191. Shards retrieved and hash-verified 12 September 2026, merged by count-weighted mean (the exit 3 merge equals the pod merge entry for entry); merged banks f2f7f7f4... (exit 0), ac73fccf... (exit 1), ed8b4990... (exit 2), aec69dba... (exit 3). Readout with the frozen scorer on the accepted confirmation states; the unembedding and final-norm tensors were fetched by byte range from the public model file at the pinned revision.